

Raed Kabir

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CS and mathematics graduate focused on backend platforms, machine learning, and high-performance software.

EDUCATION

Oregon State University

B.S. in Computer Science and Mathematics GPA: 3.7

Corvallis, OR

Sep. 2022 – June 2026

EXPERIENCE

Platform Engineer Fellow

HILOS

July 2026 – Present

Portland, OR

- Build and maintain backend services powering HILOS's **3D-printed footwear production platform** using **Django, DRF, and PostgreSQL**
- Develop **asynchronous task pipelines** with **Celery and Redis** to orchestrate production and order workflows
- Operate and iterate on services deployed to **AWS**, supporting an on-demand manufacturing pipeline

FPGA Developer Intern

Optiver

Summer 2025

Sydney, Australia

- Worked on **low-latency FPGA systems** supporting latency-critical trading infrastructure
- Developed and tested hardware components using **VHDL/SystemVerilog** and rigorous verification workflows
- Explored mechanisms for faster **network communication and data processing** across real-time trading systems

Undergraduate Research Assistant

Oregon State University

March 2024 – July 2024

Corvallis, OR

- Investigated linear algebra proofs on **matrix convexity and spectral radius behavior** with mathematics faculty
- **Developed Python scripts** to automate analysis of matrix classes and verify results computationally

Coding Instructor

Code Ninjas

Feb. 2021 – Aug. 2023

Bethany, OR

- **Hosted 10+ coding boot camps** averaging **30 students**, teaching core programming through game development
- **Designed hands-on lessons** in loops, conditionals, and events, contributing to a **30% enrollment increase**

PROJECTS

Lens Validation Tool | *C++, Qt, Qt Linguist, Git*

Fall 2025 – Spring 2026

- Built a desktop **lens validation tool** for OptiTrack's motion-capture pipeline on a student team, handed off to OptiTrack engineers
- Implemented **Chinese (Simplified) localization** via Qt's translation system, architected so new languages can be added in one place
- Built a **General settings tab** wired into the localization system, keeping language support consistent app-wide

Catalytica | *Firebase, Google Cloud, JavaScript, Tailwind CSS*

March 2025

- Led development of an AI-powered wildfire response platform for a 24-hour hackathon
- Integrated **Google Maps API** and NASA FIRMS data to visualize wildfire threats in real time
- Used **Firebase Cloud Functions** and Gemini AI to generate safety recommendations; won **1st place in the Google Tech Track**

TellTail | *Python, TypeScript, React, Tailwind CSS, PyTorch*

Jan 2025 – Feb 2025

- Trained a multi-layer neural network to **classify dog and cat breeds from images**
- Built a responsive full-stack application with a custom frontend/backend pipeline for **real-time predictions**

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, VHDL, SystemVerilog

Frameworks & Libraries: Django, Django REST Framework, React, Node.js, PyTorch, TensorFlow

Infrastructure & Tools: PostgreSQL, Celery, Redis, AWS, Docker, Firebase, Git, GDB, Neovim, Tmux

Systems & Graphics: FPGA Development, Computer Vision, OpenGL, GLSL, WebGL, Ethernet, TCP/UDP